

Horticulture Decision Making

Horticulture decision making means using information and technology to make better decisions about **fruits, vegetables, flowers, and other horticultural crops**.

Information Technology helps farmers to decide **what to grow, when to irrigate, when to apply fertilizer, how to detect diseases, and when to harvest**.

1. Crop Selection

Technology helps farmers select the **right crop and variety**.

Uses:

- Studies soil conditions.
- Check climate and weather

2. Soil Analysis

Soil data helps farmers understand the condition of the land and growth of the plant.

It provides information about:

- Soil moisture
- Soil temperature
- Soil nutrients
- Soil fertility
- Soil type

3. Irrigation Decision

Sensors and weather information help decide **when and how much water is required**.

Uses:

- Measures soil moisture.
- Provides water only when needed.
- Saves water.

4. Fertilizer Management

Technology helps farmers decide **which fertilizer to use and how much to apply**.

Uses:

- Identifies nutrient deficiency.
- Avoids excessive fertilizer use.
- Reduces fertilizer cost.

5. Disease and Pest Detection

Remote sensing, cameras, and sensors can help identify **diseases and pests**.

Uses:

- Detects affected plants.
- Finds disease at an early stage.
- Helps select suitable treatment.
- Reduces crop damage.

Example: A camera can identify unhealthy leaves and alert the farmer.

6. Weather-Based Decision Making

Weather information is very important for horticultural crops.

Farmers can use information about:

- Temperature
- Rainfall
- Humidity
- Wind
- Weather forecasts

7. Market Decision

Farmers can use digital systems to get **market information**.

Uses:

- Helps decide where to sell.
- Helps select the right time to sell.
- Improves profit.

Horticulture Decision Making Process

